

## The *data* function

The complete data for the regression task can be viewed using the *data()* function as shown here:

```
> task_mtcars$data()
  gear cyl
1:   4   6
2:   4   6
3:   4   4
4:   3   6
5:   3   8
6:   3   6
7:   3   8
8:   4   4
9:   4   4
...
```

You can also limit the number of rows and columns to be shown as illustrated below:

```
> task_mtcars$data(rows = c(1, 2, 3), cols = task_
mtcars$feature_names)
  cyl
1:   6
2:   6
3:   4
```

## Data display with *nrow*, *ncol*

The data dimensions can be displayed using *nrow* and *ncol* as shown in this snippet:

```
> c(task_mtcars$nrow, task_mtcars$ncol)
[1] 32  2
```

The features and the target column names are fetched using *feature\_names* and *target\_names*, respectively, as illustrated here:

```
> c(Features = task_mtcars$feature_names, Target = task_
mtcars$target_names)
Features  Target
  "cyl"    "gear"
```

## The *select* and *filter* functions

The *select()* function can limit the desired features to be used. In the following example, the 'cylinder' column is alone chosen from the mtcars regression.

```
> task_mtcars$select("cyl")

> task_mtcars
```

```
<TaskRegr:car> (32 x 2)
* Target: gear
* Properties: -
* Features (1):
- dbl (1): cyl
```

The rows in the data set can be selected using the *filter()* function as demonstrated below:

```
> task_mtcars$filter(1:3)

> task_mtcars$data()
  gear cyl
1:   4   6
2:   4   6
3:   4   4
```

## The *partition* function

The *partition()* function is used to split the data set into training and test data. For the given mtcars data set, we get 21 entries for the training set, and 11 data points for testing.

```
> subset = partition(task_mtcars)

> subset
$train
[1]  3  5  8  9 10 21 27 30 32  7 12 14 16 22 23 24 29 31
19 26 28

$test
[1]  1  2  4 25  6 11 13 15 17 18 20
```

## Training data

The Learner class provides many machine learning algorithms in R. In this example, we will use the *lrn()* regression tree learner for the mtcars data set and training data:

```
> learning = lrn("regr.rpart")

> learning$train(task_mtcars, row_ids = subset$train)

> learning$model
n= 21

node), split, n, deviance, yval
* denotes terminal node

1) root 21 637.86950 19.79524
 2) cyl>=5 12  74.58250 15.72500 *
 3) cyl< 5 9  99.41556 25.22222 *
```